Midterm Exam

Tuesday, March 10, 2026 4:36 PM

1. $$O_i = \frac{e^{x_i - x_{max}}}{\sum_{k=1}^{N} e^{x_k - x_{max}}}$$

• $\{x_i\} \rightarrow x_{max}$

Single Precision FP $\Rightarrow$ 4B

$e^x \rightarrow$ 10 FP ops. $\quad \sum_i \frac{x^i}{i!}$

(a) $$AI = \frac{\#\ FP\ OPs}{\#\ Bytes}$$

Operations: $\sum_{k=1}^{N} e^{x_k} \Rightarrow \underbrace{N \cdot 10}_{e^{x_k}} + \underbrace{N-1}_{sum}$

$i = 0 \sim N \quad e^{x_i} \Rightarrow \boxed{0}\ 10N$

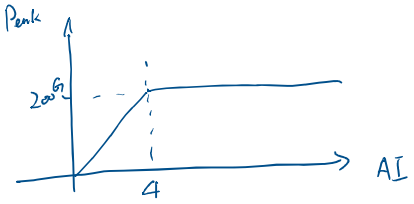
div $\Rightarrow N$

$12N - 1$

$\boxed{22N - 1}$

Bytes $\Rightarrow 4N + 4N = 8N$

(b)



$\frac{12N}{8N} = 1.5 \Rightarrow$ 75 GFLOPs

(c) Occupancy

Per thread Register usage: $20 \times 4B = 80\ B$

Per TB — Register usage: $20 \times 4B \times TB\ size = 20 \times 4 \times 128\ B$

— Share Mem: 4kB.

On each SM; 32kB RF: $32kB / (20 \times 4 \times 128) = \frac{32k}{10k} = 3$ TB

64kB share Mem. $\Rightarrow$ 64 kB / 4kB = 16 TB

On a GPU: $5 \times 3TB = 15\ TB = 15 \times 128 = 1920$

$\frac{5 \times 32kB}{80\ B}$ $\quad \frac{5 \times 32\ kB}{20 \times 4 \times 128}$

∴ SM

2. (a) `if (tid > 16)` ⇒ `s[tid] = A[tid]`

`else` `s[tid] = B[tid]`

i	s[i]
0	0
1	2
2	4
⋮	
16	32
17	35
⋮	
127	255

(b)

WO TBO

TBsize: 64

BlockDim.x

	32b			
0	0	1	2	3
1	x B000	x B004	x B008	x B00C
2	x A000	x A004	x A008	x A00C
3	x C000	x C004	x C008	x C00C
4	0	2	4	6

Predicate

0	0	0	0	0
1				
2				

16bit

0

R1 = R14 * 4

P0 = (R0 > 16)

R3 = R0 * 4 + R3

64 x 4 ↓ 256 ↓ b'100000000 ↓ x100

WO TB1

	32b			
0	64	65	66	67
1	x B001	x B000	x B000	x B000
2	x A100	x A004	x A008	x A00C
3	x C000	x C004	x C008	x C10C
4	129	131	133	135

Predicate

0	1	1	1	1
1				
2				

16bit

1

(c) Warp size 32. # SP = 16, Setp: 1 - Ld: 2 cycle. IMAD: 6cycles.

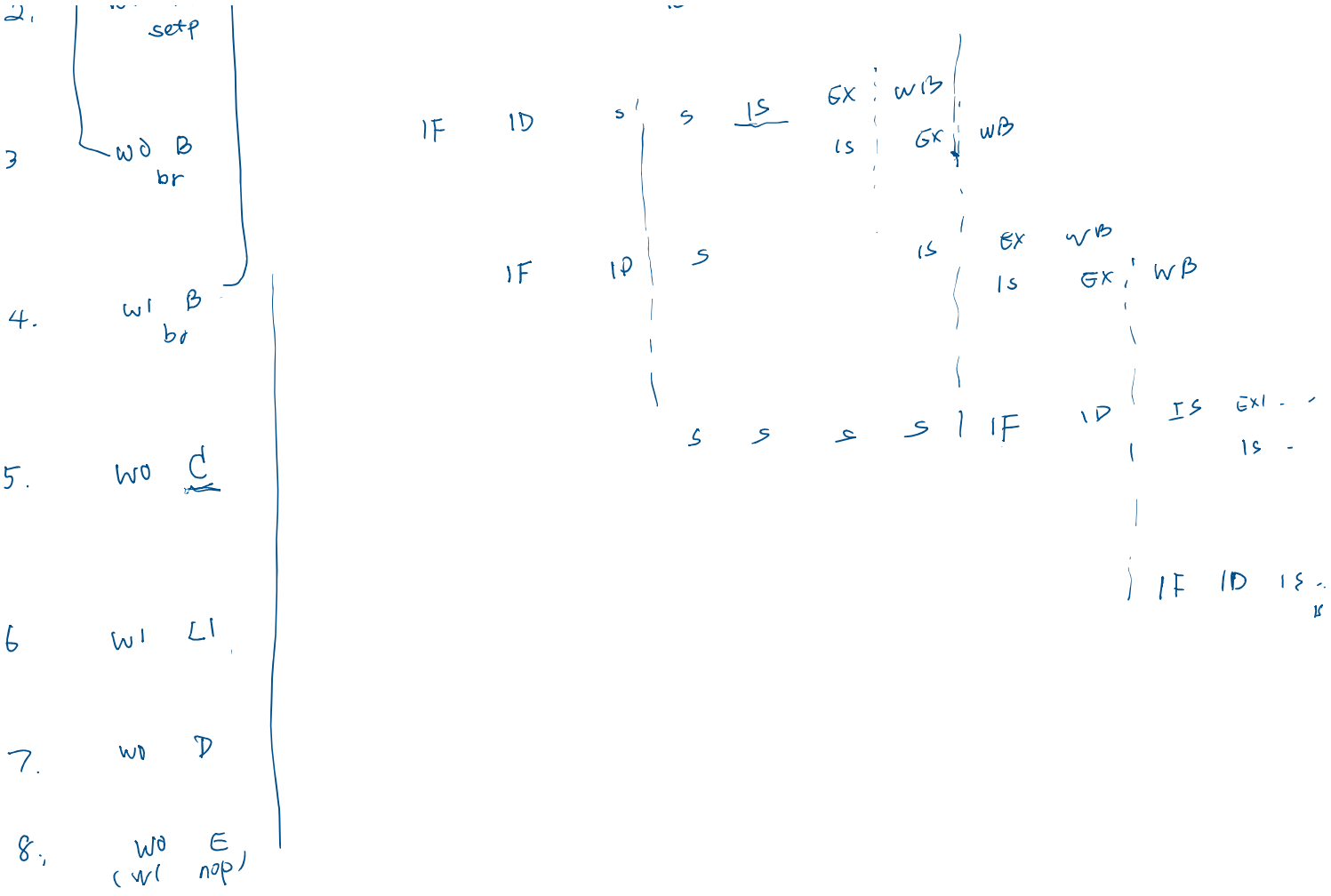
TB0_warp0_W1. on SM.

IF ID IS EX WB - RR

W0: PC = A W1: PC = A.

1. W0 A (setp) IF ID IS EX WB — IS EX WB

2. W1 A (setp) IF ID S IS EX WB — IS EX WB


2.
setp
3
w0 B
br
4.
w1 B
br
5.
w0 C
6
w1 L1
7.
w0 D
8.
w0 E
(w1 nop)
IF ID S S IS EX WB
IS EX WB
IF ID S
IS EX WB
IS EX WB
S S S S IF ID IS EX
IS
IF ID IS